

Name: _____

Teacher: _____

SYDNEY TECHNICAL HIGH SCHOOL



Mathematics Extension 1

Preliminary Assessment 1 May 2010

Time allowed – 70 minutes

Instructions

- Use a blue or black pen.
- Approved calculators may be used.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks awarded are shown on each question.
- Total marks – 55
- Attempt all questions.
- Start each question on a new page.
- Hand in your examination paper and solutions in one bundle.

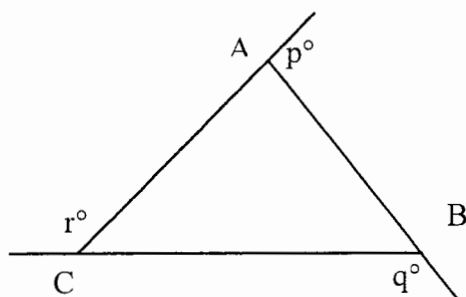
Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total

Question 1**Marks 8**

- a) Solve: $-12 \leq 1 - 2x \leq -3$. (2)
- b) Sketch the function $f(x) = \begin{cases} 1 - x, & \text{for } x < 0 \\ -1, & \text{for } x = 0 \\ 1 - x^2 & \text{for } x > 0 \end{cases}$ (2)
- c) Solve $\frac{3}{x-3} - \frac{5}{x+1} = 1$. (2)
- d) Solve by completing the square $x^2 - 4x + 1 = 0$ (2)

Question 2 (Start a new page)**Marks 10**

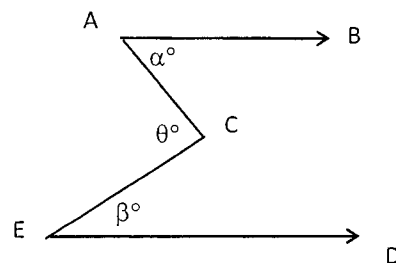
- a) Find the values x and y such that: $6 + \sqrt{x-y} = x + y + 3\sqrt{2}$. (3)
- b) For the diagram shown, prove giving reasons that $p + q + r = 360$. (2)



- c) Solve for x and show on a number line: (3)
- $$\frac{4x+1}{x-4} \geq 2$$
- d) Write down the exact value of $\operatorname{cosec} 240^\circ$, leaving denominator irrational. (2)

Question 3 (Start a new page)**Marks 9**

- a) For the function $y = \frac{1}{x^2-1}$:
- (i) State the domain. (1)
 - (ii) Find all asymptotes. (2)
 - (iii) Hence, sketch the curve. (2)
- b) Solve $|2x+1| = |x-5|$. (2)
- c) Express θ in terms of α and β , giving reasons. (2)

**Question 4** (Start a new page)**Marks 10**

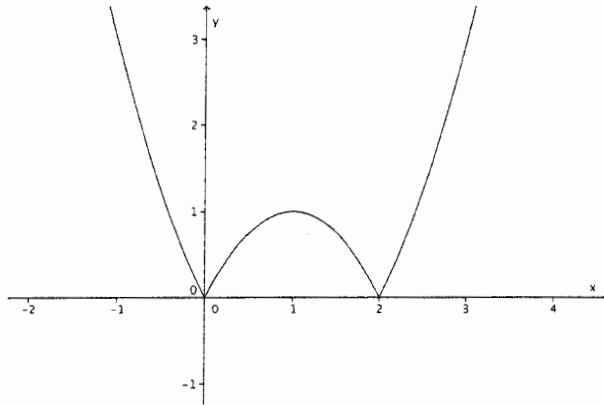
- a) Solve $2\cos 2x = \sqrt{3}$, for $0^\circ \leq x^\circ \leq 360^\circ$. (3)
- b) Solve for $0^\circ \leq \theta \leq 360^\circ$ (2)
- $\sin \theta = \cos \theta$
- c) Show that $(1 + \tan A + \sec A)(1 + \tan A - \sec A) = 2 \tan A$ (3)
- d) If $x - \frac{1}{x} = 3\sqrt{5}$, find the value of $x^2 + \frac{1}{x^2}$. (2)

Question 5 (Start a new page)

Marks 8

a) Show algebraically whether the function $f(x) = 3x(3 - x^2)$ is *odd*, *even* or neither. (2)

b) Write a possible equation for the curve shown: (2)



c) Simplify $\frac{3^{n+2} \times 12^n}{4^{n-1}}$ (2)

d) Eliminate θ from the pair of equations. (Leave in factorised form). (2)

$$x = 1 + \sec \theta$$

$$y = 2 + \tan \theta$$

Question 6 (Start a new page)

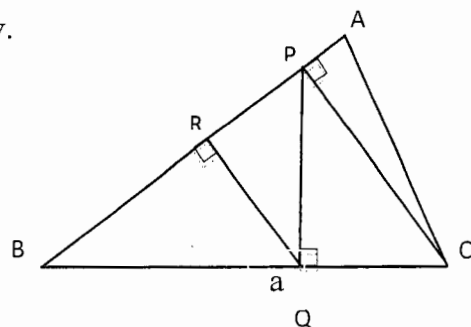
Marks 10

a) By treating the expression as a difference of two squares, express it as the product of four factors: $x^6 - y^6$. (2)

b) In triangle ABC , lines CP , PQ and QR are drawn perpendicular to AB , BC and AB respectively.

(i) Explain why $\angle RBQ = \angle RQP = \angle QPC$. (3)

(ii) Show that $QR = a \sin B \cos^2 B$. (2)



c) Simplify $\frac{1}{(a-b)(a-c)} + \frac{1}{(b-c)(b-a)} + \frac{1}{(c-a)(c-b)}$ (3)

Q1 a) Solve: $-12 \leq 1-2x \leq -3$

$$-12 \leq 1-2x \leq -3$$

$$\therefore -13 \leq -2x \leq -4$$

$$\therefore 4 \leq 2x \leq 13$$

$$\therefore 2 \leq x \leq 6.5$$

c) $\frac{3}{x-3} - \frac{5}{x+1} = 1$

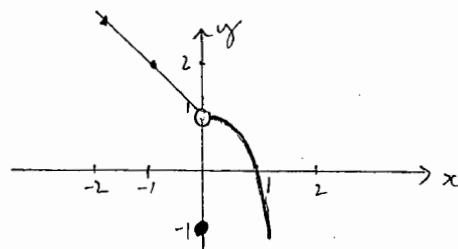
$$\therefore 3(x+1) - 5(x-3) = (x-3)(x+1)$$

$$\therefore 3x+3-5x+15 = x^2-2x-3$$

$$\therefore x^2 = 21$$

$$\therefore x = \pm\sqrt{21}$$

b)



d) $x^2 - 4x + 1 = 0$

$$\therefore (x-2)^2 - 4 + 1 = 0$$

$$\therefore (x-2)^2 = 3$$

$$\therefore x-2 = \pm\sqrt{3}$$

$$\therefore x = 2 \pm \sqrt{3}$$

Q2

a) $6 + \sqrt{x-y} = x+y + 3\sqrt{2}$
 $= x+y + \sqrt{18}$

$$\therefore x+y = 6$$

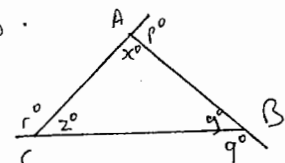
$$x-y = 18$$

$$\therefore 2x = 24$$

$$\therefore x = 12$$

$$\therefore y = -6$$

b.



Let $\hat{CAB} = x^\circ$
 $\hat{ABC} = y^\circ$
 $\hat{BCA} = z^\circ$

$$p+x = 180 \text{ (straight angle)}$$

$$q+y = 180 \text{ (" ")}$$

$$r+z = 180 \text{ (" ")}$$

$$\therefore p+x+q+y+r+z = 3 \times 180 = 540$$

$$\therefore p+q+r+x+y+z = 540$$

But $x+y+z = 180$ (angle sum of Δ)

$$\therefore p+q+r+180 = 540$$

$$\therefore p+q+r = 360 \quad \text{QED}$$

Q2 cont.

c) $\frac{4x+1}{x-4} \geq 2 \quad \text{NB } x \neq 4$

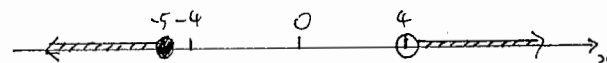
$$\therefore (4x+1)(x-4) \geq 2(x-4)^2$$

$$\therefore 4x^2 - 15x - 4 \geq 2x^2 - 16x + 32$$

$$\therefore 2x^2 + x - 36 \geq 0$$

$$\therefore (2x+9)(x-4) = 0 \quad \left[\begin{array}{l} \text{provides} \\ \text{boundaries} \end{array} \right]$$

$$\therefore x = -4.5, 4$$

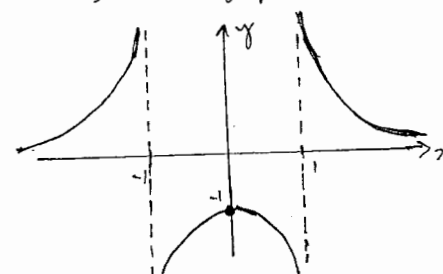


d) $\operatorname{cosec} 240^\circ = \frac{1}{\sin 240^\circ}$
 $= \frac{-1}{\sin 60^\circ}$
 $= \frac{-2}{\sqrt{3}}$
 $\left(= -\frac{2\sqrt{3}}{3} \right)$

Q3 a) $y = \frac{1}{x^2-1}$

(i) domain $\Rightarrow$ all real x except $x = \pm 1$

(ii) vertical asymptotes when $x = \pm 1$
 horizontal asymptotes when $y = 0$



As $x \rightarrow 1^-$, $y \rightarrow -\infty$

As $x \rightarrow -1^+$, $y \rightarrow -\infty$

As $x \rightarrow 1^+$, $y \rightarrow \infty$

As $x \rightarrow -1^-$, $y \rightarrow \infty$

As $x \rightarrow \infty$, $y \rightarrow 0^+$

As $x \rightarrow -\infty$, $y \rightarrow 0^+$

Q3 cont

$$b) |2x+1| = |x-5|$$

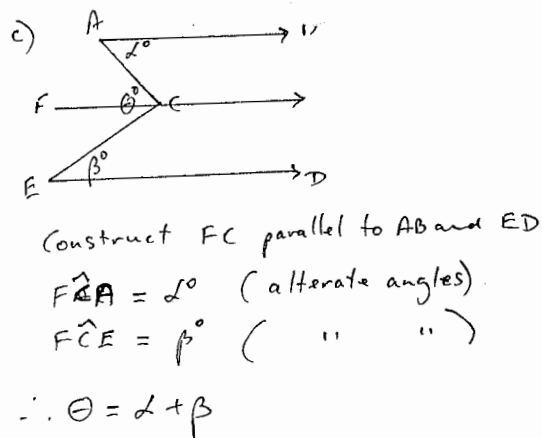
$$(2x+1)^2 = (x-5)^2$$

$$\therefore 4x^2 + 4x + 1 = x^2 - 10x + 25$$

$$\therefore 3x^2 + 14x - 24 = 0$$

$$\therefore (3x-4)(x+6) = 0$$

$$\therefore x = \frac{4}{3}, -6$$



Q4

a) $2 \cos 2x = \sqrt{3}$, for $0^\circ \leq x \leq 360^\circ$

$\therefore \cos 2x = \frac{\sqrt{3}}{2}$

$\therefore 2x = 30^\circ, 330^\circ, 390^\circ, 690^\circ$

$\therefore x = 15^\circ, 165^\circ, 195^\circ, 345^\circ$

b) $\sin \theta = \cos \theta \therefore$ first quadrant.

$\therefore \frac{\sin \theta}{\cos \theta} = 1$

$\therefore \tan \theta = 1$

$\therefore \theta = 45^\circ$ only

c) RTS $(1 + \tan A + \sec A)(1 + \tan A - \sec A) = 2 \tan A$

LHS = $(1 + \tan A + \sec A)(1 + \tan A - \sec A)$

= $(1 + \tan A)^2 - \sec^2 A$

= $1 + 2 \tan A + \tan^2 A - \sec^2 A$

= $2 \tan A + \sec^2 A - \sec^2 A$

= $2 \tan A$

= RHS QED

d) $x - \frac{1}{x} = 3\sqrt{5}$

$\therefore (x - \frac{1}{x})^2 = 45$

$\therefore x^2 - 2 + \frac{1}{x^2} = 45$

$\therefore x^2 + \frac{1}{x^2} = 47$

Q5

a)

$f(x) = 3x(3-x^2)$

$f(-x) = -3x(3-(-x)^2)$

= $-3x(3-x^2)$

= $-f(x)$

$\therefore$ the function is odd.

d) $x = 1 + \sec \theta$

$y = 2 + \tan \theta$

$\therefore \sec \theta = x - 1$ (i)

$\tan \theta = y - 2$ (ii)

Now $\sec^2 \theta = 1 + \tan^2 \theta$

$\therefore (x-1)^2 = 1 + (y-2)^2$

b) $y = |x(x-2)|$

= $|x^2 - 2x|$

c) $3^{\frac{n+2}{4^{n-1}}} \times 12^n = \frac{3^{n+2} \times 3^n \times 4^n}{4^{n-1}}$

= $3^{2n+2} \times 4$

= $4 \times 3^{2(n+1)}$

Q6

a) $x^6 - y^6 = (x^3 - y^3)(x^3 + y^3)$

= $(x-y)(x^2 + xy + y^2)(x+y)(x^2 - xy - y^2)$

b) (i) $\widehat{RQB} = 90 - \widehat{RBQ}$ (right-angled Δ)

$\therefore \widehat{RAP} = 90 - (90 - \widehat{RBQ})$ (complementary angles)

= $\widehat{RBQ}$

Now $\widehat{RPA} = 90 - \widehat{RAP}$

So $\widehat{CPQ} = 90 - (90 - \widehat{RPA})$ (complementary angles)

= $\widehat{RPA}$

= $\widehat{RBQ}$

$\therefore \widehat{RBQ} = \widehat{RAP} = \widehat{CPQ}$ QED

(ii) $PC = a \sin B$ in ΔPBC

$\therefore \frac{PQ}{a \sin B} = \cos B$ in ΔPQC

$\therefore PQ = a \sin B \cos B$

$\therefore \frac{RQ}{a \sin B \cos B} = \cos B$ in ΔRQP

$\therefore RQ = a \sin B \cos^2 B$ QED

c) $\frac{1}{(a-b)(a-c)} + \frac{1}{(b-c)(b-a)} + \frac{1}{(c-a)(c-b)}$

= $\frac{(b-c) - (a-c) + (a-b)}{(a-b)(b-c)(a-c)}$

= $\frac{b-c-a+c+a-b}{(a-b)(b-c)(a-c)}$

= 0